Tiago Autonomous Exploration and Manipulation

Autonomous and Mobile Robotics - Project

Francesco Cenni Giulio Mazzaroppi Giacomo Pezzolati

francesco.cenni3@studio.unibo.it
giulio.mazzaroppi@studio.unibo.it
giacomo.pezzolati@studio.unibo.it

University of Bologna

Academic Year 2025/2026

February 7, 2026

Overview

1. Introduction
2. Task 1: Exploration
3. Task 2: Navigation
4. Task 3: Manipulation

Project Overview

- **Objective:** Develop a complete autonomous system for the TiaGo robot combining navigation and manipulation skills.
- **Tasks:**
  1. Map Generation (SLAM).
  2. Autonomous Localization & Exploration.
  3. Pick and Place of specific ArUco cubes.



Figure 1: Map Generation in RViz.



Figure 2: Autonomous Localization & Exploration



Figure 3: Pick and Place of specific ArUco cubes

System Architecture & Design

Development Stack

- **Framework:** ROS2 Humble Hawksbill.
- **Simulation:** Gazebo 11 (Physics & Environment).
- **Visualization:** RViz2 (Real-time sensor data & markers).

Architectural Pattern

We adopted a modular **Policy vs. Actuator** approach:

- **Main Policy (Coordinators):** High-level nodes managing the task logic and state machine.
- **Generalized Actuators:** Reusable skills (e.g., `map_navigator`, `head_motion`) agnostic of the specific task.
- **Handshake Protocol:** Coordination is handled via strictly defined Topics, Services, and Actions to ensure robust synchronization[a].

[a] Custom messages and service definitions are isolated in a dedicated **ament_cmake** package: `tiago_exam_interfaces`.

Package Organization: `tiago_exam`

The Project is entirely accomplished inside the `tiago_exam` package. This is a **ament_python** package structured as follows:

- **launch:** Contains three launch files to start each respective task environments.
- **tiago_exam:** Organized into sub-packages for each task and a shared utilities package.
- **data:** Contains the csv file for the aruco markers records used in task2 and task3.
- **config:** Stores RViz configuration files and map data for easy access across tasks.
- **temp:** Contains the debug map images for the exploration surface and reference frames.

Task 1: Autonomous Map Generation

Objective

Perform a full autonomous exploration of an unknown environment to generate a consistent 2D occupancy grid map using SLAM.

Environmental Challenges

The testing environment presents specific difficulties that standard configuration could not handle:

- **Complex Geometry:** Narrow passages and obstacles requiring precise maneuvering.
- **Lighting Artifacts:** Strong shadows created false positive obstacles in the LIDAR readings (ghost objects).



Figure 4: Environment in Gazebo 11.

Task 1: Exploration Strategy

To achieve efficient coverage, we integrated a C++ node based on `m-explore`, enhancing the standard frontier exploration logic.

Optimized Frontier Selection Logic

Instead of a greedy approach, we tuned the cost function parameters to prioritize frontiers that are:

1. **Closer** to the robot's current pose (minimizing travel time).
2. Located in **larger unexplored voids** (maximizing information gain).

Completion Workflow:
Upon exhaustion of the whitelist frontiers, the robot automatically navigates back to the **Home Position** and triggers the map saver, storing the result in the `maps/` directory.

Task 1: System Configuration & Safety

Safety Measure: Pre-Exploration Configuration

Before moving, the system executes a mandatory **Tuck Arm** procedure. Folding the arm into a safe configuration minimizes the robot's footprint, preventing collisions with low obstacles or door frames during autonomous navigation.

Sensor & Navigation Tuning

We refined the navigation stack parameters to address the environmental noise:

- **LIDAR Raytracing:** Adjusted strictly to filter out shadow reflections and prevent "ghost obstacles" on the map.
- **Motion Profile:** Smoothed acceleration limits to ensure stable mapping and prevent odometry drift.
- **Cost Map Parameters:**



Figure 5: Generated Occupancy Grid

Task 1: Exploration Video Result



Figure 6: Video Placeholder - Link to YouTube or local file.

Task 2: System Architecture & Orchestration

The complex behavior required for Task 2 is managed by a central **Coordinator Node** that acts as the high-level decision maker.

The Coordinator Design

- **Finite State Machine (FSM):** Strict transition logic between *Exploring*, *Tracking*, and *Inspecting* states.
- **Concurrency Handling:** Utilizes `ReentrantCallbackGroup` and `MultiThreadedExecutor`. This allows the node to process sensory interrupts (e.g., seeing a marker) immediately, even while blocking navigation actions are active.
- **Interrupt Logic:** The standard exploration loop is designed to be preempted instantly upon ArUco detection.



Figure 7: State Machine Logic

Task 2: Global Localization

Since the robot spawns in an unknown location, standard AMCL localization is insufficient immediately after startup. We implemented a robust initialization routine.

The Converge Strategy:

1. **Particle Spread:** Trigger `/reinitialize_global_localization` to distribute AMCL particles uniformly across the map.
2. **Data Gathering:** The robot executes a 360° in-place rotation at 1.0 rad/s to feed the laser scan matcher.
3. **Covariance Check:** The routine continuously monitors the pose covariance matrix, ensuring it falls below strict thresholds before allowing any movement.



Figure 8: AMCL Particle Convergence



Figure 9: Covariance Reduction Over Time

Task 2: Exploration Strategy

After the localization, the Coordinator Node activate the following nodes:

Generalized Nodes

- **Aruco Marker Detector:** The node continuously monitors the camera feed for the station markers.
- **Explorator:** The node processes the occupancy grid map to create a "whitelist" of safe navigation targets, ensuring the robot only attempts to navigate to locations that are free of obstacles.
- **Navigator:** The robot continuously samples from the "whitelist" of safe exploration goals. After reaching each goal, the robot performs a 360° rotation to improve its environmental awareness.



Figure 10: Clearance-Based Goal Sampling



Figure 11: Random Safe Goal Navigation

Task 2: Exploration Strategy - Contd.

Exploration Strategy

- **Tracking:** Once an ArUco marker is detected, the robot is brought to a specific distance from the marker, while continuously facing it by adjusting the camera tilt and pan angles.
- **Inspecting:** Once the robot is correctly positioned, it transitions to the Inspecting state, where the marker and robot Pose is saved in a csv file for later use.
- **Homing:** After all the markers are detected, the robot automatically navigates back to the Home Position.



Figure 12: Aruco Tracking Behavior



Figure 13: Aruco Inspecting Behavior

Task 2: Safety Navigation

Map Processing Pipeline

Using OpenCV on the occupancy grid map:

1. **Safety Mapping:** Applied `cv2.distanceTransform` to calculate the distance of every pixel to the nearest obstacle.
2. **Filtering:** Created a "Whitelist Mask" of pixels where:

$$d_{obstacle} > (R_{robot} + \text{Clearance}_{0.7m})$$

3. **Goal Selection:** Randomly samples coordinates from the whitelist and converts them from Image Frame (pixels) to World Frame (meters).

Result: The robot never attempts to navigate into walls or narrow traps.

Task 2: Navigation Video Result



Figure 14: Video Placeholder - Link to YouTube or local file.

Task 3: Manipulation Coordination

The `coordinator_task3` node orchestrates the sequence, bridging the data gathered in Task 2 with the actuation logic.

Execution Workflow

1. **Data Ingestion:** Parses `marker_locations.csv` to retrieve the exact pose of the targets found during exploration.
2. **State Sequence:** Enforces the required order of operations:

$$\text{Localize} \xrightarrow{\text{Navigate to Pick}} \text{Pick Cube ID 63} \xrightarrow{\text{Navigate to Place}} \text{Place Cube ID 63}$$

$$\text{Next Target} \xrightarrow{\text{Navigate to Pick}} \text{Pick Cube ID 582} \xrightarrow{\text{Navigate to Place}} \text{Place Cube ID 582}$$

3. **Visual Feedback:** Publishes dynamic TF frames (`target_cube_frame`) to visualize the manipulation target in RViz in real-time.

Task 3: Grasping & Manipulation Strategy

The manipulation pipeline implements a State Machine handling

- **Grasp Procedure:** *Pre-Grasp* $\rightarrow$ *Grasp* $\rightarrow$ *Post-Grasp*.
- **Release Procedure:** *Pre-Release* $\rightarrow$ *Release* $\rightarrow$ *Post-Release*.

Between this two main phases the robot perform navigation (handled by the coordinator and the navigation stack). To ensure a **safe** navigation the manipulator also perform additional phases.

- **Safe Pose:** Before navigating, the arm moves to a predefined "safe" configuration to avoid collisions with the environment during navigation.
- **Idle:** After moving to the safe pose, the arm enters an idle state, waiting for the coordinator to signal the next command.

Task 3: Grasp & Release Selection Pose

Pose Calculation

The Grasp Pose is computed by extracting the marker's transform. The choice of the grasping point has been made to ensure a stable grip on the cube, considering collinear and antipodal grasp for the 7cm cube.

Pose Computation (KDL)

Given the marker pose T_{base}^{marker}, we compute the approach vector by applying a geometrical offset:

$$T_{grasp} = T_{base}^{marker} \cdot T_{rotY(90°)} \cdot T_{offset(-0.4m)}$$

This aligns the gripper normal with the cube's face.



Figure 15: TF Frames visualization

Placement Logic: The release pose is derived from the grasp pose but lifted vertically to ensure the object rests on the table:

$$P_{release}.z = P_{grasp}.z + 0.15m$$

Task 3: Precise Placement Pose

The placement pose is chosen to be the same as the grasp pose to ensure that the cube is placed in approximately the same position as the one it was picked from. This is because the robot has to place the cube in a specific area of the table, and using the same pose for both picking and placing allows for a more consistent and accurate placement.
The grasp pose is saved and re-used for the placement phase, for this reason, an exact distance between the robot and the table is required to ensure a successful grasp and placement.
To achieve this, once the robot reaches the grasp pose in the map, it measures the distance to the table and record it. This distance is then used to adjust the robot's position during the placement phase.

Precise Approach Logic

To compensate for navigation errors, the node reads the real-time distance from the table. If the robot is wrongly positioned ($> 1cm$ error), it triggers a closed-loop **Linear Approach/Backup** maneuver before attempting the grasp.

Simulation Stability: Link Attacher

Physics engines (Gazebo/ODE) struggle with friction simulation during mobile manipulation, causing objects to slip from the gripper during base motion.

The Problem

After modifing the gripper and object frictions and compliance properties, we tried to implement a force grasp strategy by subscribing to the gripper's FT Sensor data. The gripper would close until a the 10N threshold was reached in the specific axis.

The Solution

We integrated the **Link Attacher Plugin** to enforce a rigid kinematic link between the robot and the object during transport.

Implementation Cycle:

1. **Grasp:** Gripper closes physically on the object.
2. **Lock:** Call `/ATTACHLINK` service $\rightarrow$ Object becomes part of the robot's TF tree.
3. **Transport:** Robot navigates to the place station.
4. **Release:** Gripper opens $\rightarrow$ Call `/DETACHLINK` service $\rightarrow$ Physics are restored